

RULES AND REGULATIONS FOR THE WRITTEN EXAMS
AT THE FACULTY OF ENGINEERING AND NATURAL SCIENCES

Please read this document carefully, sign the bottom part, and include this form with your solution sheet.

- The exam duration (start-end times) will be written on your exam sheet or will be announced, you should adhere to these times.
- You cannot leave the examination room in the **first 30 minutes** of the exam.
- You are **NOT allowed** to take the exam once a student has left one of the exam rooms.
- Talking with another student during the exam for any reason (asking for an explanation, for a calculator or eraser or any other object) is **NOT allowed**.
- If you **GIVE** to or **RECEIVE** from a student any item (eraser, calculator, pen, etc.) during the exam, this will be taken as a sign of cheating.
- **TURN OFF** your CELLPHONES AND SMART DEVICES; and **PLACE THEM away from you** such that those **CANNOT BE SEEN** from outside.
- **HAVING a CELLPHONE or a SMART DEVICE** that is **VISIBLE FROM OUTSIDE** during the exam is **NOT ALLOWED**. Interacting with a cellphone in any way (EVEN IF IT IS TURNED OFF) will be taken as a sign of cheating.

In case of a sign of cheating, or if you do not obey the above rules, there will be a disciplinary action taken towards you based on the STUDENT DISCIPLINARY REGULATIONS of YÖK Items 2.5, 2.7, and 2.8.

I understand the above-mentioned rules and regulations, and I hereby accept them, and any consequences following my actions during the exam.

Student Full Name:

Student Number:

Course Code and Title:

AIN-2002 INTRODUCTION TO DATA SCIENCE

Signature:

Date:

JUNE 29th, 2024

Q.1	Q.2	Q.3	Q.4	Q.5	TOTAL
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Q.1) (10) Write which of the following is an example of structured data, unstructured data, or semi-structured data.

A JSON file	
A CSV file	
A tweet	
An email	

Q.2) (30) You are working as a data scientist for a healthcare company. The company has collected a dataset containing various features of patients, such as age, blood pressure, cholesterol levels, and other medical history details. The goal is to predict whether a patient is at risk of developing heart disease (a binary outcome: "Yes" or "No").

Patient ID	Age	Gender	Blood Pressure	Cholesterol	Smoking	Physical Activity	BMI	Diabetes	Family History	Heart Disease
1	45	M	130	200	No	Yes	25.3	No	Yes	No
2	50	F	120	210	Yes	No	27.6	Yes	Yes	Yes
3	39	M	140	180	No	Yes	22.9	No	No	No
4	55	F	150	230	Yes	No	30.1	Yes	Yes	Yes
5	60	M	135	195	Yes	Yes	28.4	No	Yes	Yes
6	52	F	110	205	No	Yes	24.2	Yes	No	No
7	47	M	145	220	Yes	No	26.7	No	Yes	Yes
8	48	F	125	190	No	Yes	23.5	No	No	No
9	63	M	160	240	Yes	No	29.8	Yes	Yes	Yes
10	40	F	118	185	No	Yes	21.9	No	No	No

Column Descriptions:

PatientID: Unique identifier for each patient.

Age: Age of the patient.

Gender: Gender of the patient (M for Male, F for Female).

BloodPressure: Blood pressure measurement of the patient.

Cholesterol: Cholesterol level of the patient.

Smoking: Indicates if the patient is a smoker (Yes or No).

PhysicalActivity: Indicates if the patient engages in regular physical activity (Yes or No).

BMI: Body Mass Index of the patient.

Diabetes: Indicates if the patient has diabetes (Yes or No).

FamilyHistory: Indicates if there is a family history of heart disease (Yes or No).

HeartDisease: Target variable indicating if the patient is at risk of developing heart disease (Yes or No).

The sample data is provided to help you better understand the problem.

Please, do NOT develop a solution that is specific to the sample data provided above!

Based on the given case study, which ***supervised learning algorithm*** would be most suitable for predicting whether a patient is at risk of developing heart disease? Explain why the chosen algorithm is appropriate for this prediction task. Include the key characteristics of the algorithm and how it fits the nature of the problem.

Q.3) (20) You are a data scientist for a network security company. The company wants to detect unusual patterns in network traffic that could indicate security breaches or attacks. The dataset includes features such as source IP, destination IP, protocol, bytes transferred, and connection duration:

ConnectionID	SourceIP	DestinationIP	Protocol	Bytes Transferred	Connection Duration
1	192.168.1.1	192.168.1.100	TCP	1500	30
2	192.168.1.2	192.168.1.101	UDP	500	10
3	192.168.1.3	192.168.1.102	TCP	10000	300
4	192.168.1.4	192.168.1.103	UDP	200	5
5	192.168.1.1	192.168.1.100	TCP	2500	60
6	192.168.1.2	192.168.1.101	UDP	150	8
7	192.168.1.3	192.168.1.102	TCP	12000	400
8	192.168.1.4	192.168.1.103	UDP	180	4
9	192.168.1.1	192.168.1.100	TCP	300	15
10	192.168.1.2	192.168.1.101	UDP	1000	25
11	192.168.1.3	192.168.1.102	TCP	15000	500
12	192.168.1.4	192.168.1.103	UDP	50	2

Column Descriptions:

ConnectionID: Unique identifier for each connection.

SourceIP: IP address of the source.

DestinationIP: IP address of the destination.

Protocol: Protocol used for the connection (e.g., TCP, UDP).

BytesTransferred: Number of bytes transferred during the connection.

ConnectionDuration: Duration of the connection in seconds.

The sample data is provided to help you better understand the problem.

Please, do NOT develop a solution that is specific to the sample data provided above!

(This is a 100-minute exam)

Given the following sample network traffic data, develop a strategy to detect anomalies. Choose an appropriate anomaly detection algorithm, justify your choice, and describe the steps you would take to implement it. Discuss how you would evaluate the performance of your anomaly detection model.

Q.4) (20) You are a data scientist for an online retail company specializing in electronics and gadgets. The company wants to leverage customer purchase data to improve marketing strategies and personalize the shopping experience. The dataset includes features such as customer ID, product categories purchased, purchase frequency, and total spending:

CustomerID	Category1	Category2	Category3	Category4	Frequency	TotalSpending
1	Electronics	Books		Sports	5	1500.00
2	Electronics		Kitchen		3	800.00
3	Books	Electronics		Toys	7	2500.00
4	Electronics	Toys		Sports	4	1200.00
5	Kitchen	Electronics		Toys	6	1800.00
6	Electronics		Sports		2	500.00
7	Books	Toys		Electronics	8	3000.00
8	Electronics		Books	Toys	5	2000.00
9	Electronics	Sports		Books	3	1000.00
10	Sports	Books		Electronics	4	1500.00

The sample data is provided to help you better understand the problem.

Please, do NOT develop a solution that is specific to the sample data provided above!

Develop a recommendation system to suggest new products to customers based on their segment. Describe how you would implement this recommendation system.

MAKEUP EXAM

(This is a 100-minute exam)

Q.5) (20) A retail store has provided you with a sample of their daily sales data for the past week in a CSV file named "sales_data.csv". The data includes columns such as date, product category (category), product ID (product_id), quantity sold (quantity), and total sales amount (sales_amount).

Sample Data (sales_data.csv):

```
date,category,product_id,quantity,sales_amount
2024-05-20,Electronics, TV123, 2, 1200.00
2024-05-20,Clothing, Shirt456, 1, 35.99
2024-05-21,Appliances, Fridge789, 1, 899.99
2024-05-21,Grocery, Milk123, 3, 5.99
2024-05-22,Electronics, Monitor007, 1, 250.00
2024-05-22,Clothing, Pants987, 2, 79.98
2024-05-23,Appliances, Washer345, 2, 599.98
2024-05-23,Grocery, Eggs567, 1, 2.99
```

a) (10) Calculate the average sales amount per day.

```
import pandas as pd

sales_df = pd.read_csv("sales.csv")
```

b) (10) Filter and display all sales records for the 'Electronics' category.

```
import pandas as pd

sales_df = pd.read_csv("sales.csv")
```